

Product Vision Document

Design and Architecture of an End-to-End Event Ticketing Platform

Document Information

Item	Description
Project Title	Design and Architecture of an End-to-End Event Ticketing Platform
Document Type	Product Vision Document
Course	Software Engineering II
System Type	Web-based event ticketing platform
Main Focus	Event discovery, real-time seat selection, temporary seat locking, secure payment, QR-code ticket generation, notifications, and scalable architecture

1. Introduction

This document describes the product vision for an end-to-end event ticketing platform. The main goal of this platform is to allow users to search for different events, select their preferred seats online in real time, reserve tickets, complete payment, and finally receive their tickets with a QR Code.

The system is not designed only for ticket buyers. It also provides separate features for event organizers and system administrators. Event organizers can create events, define venues and seat layouts, set ticket prices, and manage ticket sales. System administrators are responsible for managing users, monitoring system performance, controlling platform security, and managing the overall system.

This product is intended to behave like a realistic ticketing platform that supports the full ticket purchase process from event discovery to final ticket validation.

2. Product Vision Statement

The vision of this product is to create a modern, reliable, scalable, and secure platform for selling event tickets online. The platform should make the ticket purchasing process simple, fast, transparent, and error-free for users.

One of the most important goals of the system is to remain stable during high-demand events, even when many users enter the system at the same time. The system must ensure that no two users can purchase the same seat for the same event.

For this reason, double-booking prevention, high-traffic management, temporary seat locking, secure payment processing, and scalable architecture are core parts of the product vision.

3. Target Users

3.1 Regular Users / Ticket Buyers

Regular users are people who want to browse events and buy tickets online. They need a simple and fast interface that allows them to search for events, view available seats, select seats, complete payment, and receive their final tickets.

Main Needs of Regular Users

- Search and filter events.
- View event details.
- View venue layout and available seats.
- Select seats in real time.

- Temporarily reserve selected seats.
 - Complete secure payment.
 - Receive a ticket with QR Code.
 - Receive confirmation through SMS or email.
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3.2 Event Organizers

Event organizers are individuals or organizations that want to register and manage their events on the platform. They must be able to manage event information, venue details, seat capacity, ticket prices, and sales status.

Main Needs of Event Organizers

- Create and edit events.
 - Define venues and seat layouts.
 - Set ticket prices.
 - Manage event capacity and schedule.
 - View ticket sales.
 - View revenue and reports.
 - Manage event status.
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3.3 System Administrators

System administrators are responsible for managing the overall platform. They must be able to control users, event organizers, roles, system security, and the general status of platform services.

Main Needs of System Administrators

- Manage users and roles.
 - Review and manage event organizers.
 - Monitor system performance.
 - Manage errors and incidents.
 - Control system security.
 - Check service status.
 - Handle possible issues reported by users and organizers.
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4. Main Product Goals

4.1 Simple and Smooth User Experience

The system should be designed in a way that allows users to find their desired events and complete the ticket purchasing process without confusion. The user journey from event search to ticket delivery must be clear, fast, and easy to understand.

4.2 Real-Time Seat Selection

Users should be able to view the venue seat map and see the status of seats in real time. Seats should be shown with different states such as available, locked, and reserved so that users can make correct decisions while selecting seats.

4.3 Prevention of Simultaneous Seat Reservation

One of the most important parts of this project is preventing double-booking. This means the system must prevent two users from purchasing the same seat at the same time.

To achieve this, the system uses a temporary seat locking mechanism. When a user selects a seat, that seat is locked for a limited time so the user can complete the payment. During this time, other users cannot select the same seat.

4.4 Secure Payment and Final Ticket Issuance

After selecting seats, the user proceeds to the payment step. If the payment is successful, the temporary reservation is converted into a final purchase, and the system issues the ticket.

Each ticket must contain a unique QR Code that can be used for ticket validation.

If the payment fails or the user does not complete the payment within the allowed time, the locked seat must be released automatically so that other users can select it.

4.5 Scalability During High Traffic

For popular events, many users may enter the system at the same time. Therefore, the system must be able to handle high traffic without failure.

To support this goal, the system can use:

- Virtual waiting room.
- API Gateway.
- Rate limiting.

- Caching.
 - Queue-based processing.
 - Scalable and decoupled architecture.
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4.6 Decoupled and Maintainable Architecture

The system should be divided into separate components or services so that development, testing, maintenance, and scaling become easier.

The main parts of the system include:

- Authentication service.
 - Event catalog service.
 - Seat reservation service.
 - Payment service.
 - Notification service.
 - Reporting and monitoring service.
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5. Main System Features

The main features of the platform include:

- User registration and login.
- Secure authentication.
- Role-based access control.
- Event creation and management.
- Venue and seat layout management.
- Event search and filtering.
- Real-time seat status display.
- Temporary seat locking.
- Reservation management.
- Payment processing.
- Ticket issuance with QR Code.
- SMS or email confirmation.
- Virtual waiting room.
- Organizer dashboard.

- System admin dashboard.
 - Reporting and monitoring.
 - Error and incident management.
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6. Business Value of the Product

This platform makes the event ticket selling process simpler, more reliable, and more transparent. In traditional methods or weak systems, problems such as selling the same seat to multiple users, payment errors, lack of accurate reports, and system overload during popular events may happen.

This system reduces these problems by using proper architecture, temporary seat locking, secure payment processing, and fast notifications.

The platform increases trust for both users and event organizers. Users can buy tickets with more confidence, while organizers can manage ticket sales more effectively and access accurate information about event status and revenue.

7. Product Success Criteria

The success of this system can be evaluated using the following criteria:

- Number of successful ticket purchases.
 - Payment success rate.
 - Reduction or elimination of double-booking errors.
 - Average system response time.
 - System stability during high-traffic events.
 - Number of tickets issued without error.
 - User satisfaction level.
 - Organizer satisfaction level.
 - Success rate of SMS and email notifications.
 - System availability.
 - Reduction of reservation and payment-related errors.
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8. Project Assumptions

The following assumptions are considered in the design of this system:

- Users access the system through the internet.

- The system is designed as a web-based platform.
 - Payments are processed through an external payment gateway.
 - PostgreSQL is used as the main database.
 - Redis is used for temporary seat locking.
 - RabbitMQ or Kafka is used for asynchronous communication.
 - Docker is used for local service execution.
 - Kubernetes is considered for demonstrating scalable deployment architecture.
 - Terraform is used for demonstrating infrastructure provisioning.
 - The system should be extendable for future versions.
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9. Future Scope of the Product

In future versions, the following features can be added to the system:

- Mobile application.
 - Dynamic pricing based on demand.
 - Ticket cancellation and refund management.
 - Ticket transfer between users.
 - Fraud detection system.
 - Integration with multiple payment gateways.
 - Event recommendation system.
 - Advanced sales analytics for organizers.
 - Multi-language support.
 - Support for both online and in-person events.
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10. Conclusion

The End-to-End Event Ticketing Platform is a complete system for managing the ticket purchasing process from start to finish. It begins with event discovery and seat selection and continues through secure payment, ticket issuance, notification, and event management.

The main focus of this project is to design a reliable architecture, prevent simultaneous reservation of the same seat, manage high traffic, use decoupled services, and prepare the system for future scalability.

By using PostgreSQL, Redis, an asynchronous messaging system such as RabbitMQ or Kafka, Docker, Kubernetes, and Terraform, this project can be presented as a complete, realistic, and defensible Software Engineering II system design project.